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CCN Experiment 7

SSL/TLS Security Assessment of Web Servers

AIM

To use an online SSL/TLS assessment tool to determine the security characteristics of web servers, and to examine how the server-side configuration shapes browser/server handshake behaviour.

PROBLEM STATEMENT

Assess six public web servers across three sectors — Education, Banking and Trading — using the Qualys SSL Server Test. For every server, record the overall grade, the four Summary metrics (Certificate, Protocol Support, Key Exchange, Cipher Strength) and the visible notices for TLS 1.3, post-quantum cryptography (PQC) key exchange and HTTP Strict Transport Security (HSTS). Compare the two servers within each sector, contrast the sectors, and relate the findings to TLS handshake security.

REQUIREMENTS

A computer with Internet access, a modern web browser, and the Qualys SSL Server Test (https://www.ssllabs.com/ssltest/). Only passive, publicly available assessments are used; no server is probed beyond the standard test or modified.

THEORY

SSL/TLS and the handshake

SSL/TLS secures HTTP traffic by authenticating the server and agreeing on session keys that provide confidentiality and integrity. "SSL" is the historic name; modern HTTPS uses TLS. A TLS handshake proceeds roughly as follows:

1. The client (browser) sends a ClientHello listing the TLS versions, cipher suites, key-share data and extensions it supports.
2. The server replies with a ServerHello choosing mutually supported parameters.
3. The server proves its identity with an X.509 certificate and shows possession of the matching private key.
4. Both sides derive the traffic keys and exchange Finished messages; application data is then encrypted.

TLS 1.3 shortens this exchange and removes legacy algorithms, so a server that supports it exposes a smaller attack surface than one that still offers TLS 1.0/1.1.

Qualys SSL Server Test

The Summary page gives an overall letter grade (A+ to F) plus four scores out of 100: Certificate, Protocol Support, Key Exchange and Cipher Strength. The grade is not a plain average of these scores — configuration issues can cap it (for example, still supporting TLS 1.0/1.1 limits the grade to B), and good practice such as HSTS can lift it. A higher grade indicates a stronger assessed configuration, not a guarantee that a site has no vulnerabilities.

Related terms

- **HSTS:** tells browsers to use only HTTPS for future visits, defeating protocol-downgrade and SSL-stripping attacks.

- **SNI (Server Name Indication):** lets the client name the host it wants during the handshake, so one IP address can serve many certificates; very old clients without SNI cannot connect.
- **PQC / hybrid key exchange:** combines a classical exchange with a quantum-resistant one to protect recorded traffic against future quantum computers. Its absence is not a present weakness.
- **Limits of a summary:** cipher-suite order, certificate chain details and handshake simulation sit in the detailed report sections, not in the Summary screenshots.

PROCEDURE

1. Open the Qualys SSL Server Test at https://www.ssllabs.com/ssltest/.
2. Enter each selected public hostname and view its report; where several endpoints are listed, open one specific endpoint.
3. Under Summary, note the overall grade, the four metric bars and any TLS 1.3, PQC, HSTS or SNI notices.
4. Scan additional hosts in the Education and Banking sectors and pick the best- and worst-rated host of each sector for the comparison; both Trading hosts were scanned directly.
5. Capture the Summary screenshot of each selected host and keep two screenshots per sector in the report.
6. Compare the two hosts within each sector, then compare the sectors with one another.
7. Restrict the activity to passive assessment of public endpoints.

WEBSITES SELECTED

Sector	Website	Hostname	Endpoint tested	Assessed (UTC)
Education	IIM Bangalore	www.iimb.ac.in	13.233.203.120	16 Oct 2026, 03:46
Education	SPIT Mumbai	www.spit.ac.in	43.252.193.19	16 Oct 2026, 03:20
Banking	Axis Bank	www.axis.bank.in	23.37.18.32	Not shown
Banking	Janata Sahakari Bank (JSB), Pune	jsblpune.bank.in	82.165.3.153	16 Oct 2026, 03:59
Trading	Upstox	upstox.com	18.238.96.74	Not shown
Trading	Groww	groww.in	2a06:98c1:3100::6812:25d6	Not shown

Source of observations: the Qualys SSL Labs Summary screenshots of the six hosts above. "Not shown" means the screenshot has no assessment timestamp.

Selection basis

Within Education and Banking, more hosts were scanned and the best- and worst-rated host was taken for the comparison. Education: IIM Bangalore (A+) is the best and SPIT (B) the weakest of three scans. Banking: Axis Bank and SBI both scored A+, but Axis was chosen because it also reports PQC support; JSB Pune (B) is the weakest of four scans. In Trading, both hosts were scanned directly.

Other host scanned (screenshot omitted)	Sector	Grade	Visible notice
www.iitm.ac.in	Education	A−	HSTS invalid/disabled; no PQC

Other host scanned (screenshot omitted)	Sector	Grade	Visible notice
onlinesbi.sbi.bank.in	Banking	A+	TLS 1.3 and long HSTS; no PQC
pnb.bank.in	Banking	A	TLS 1.3, long HSTS, CAA; no PQC

OBSERVATIONS — CONSOLIDATED COMPARISON

Table 1. Grades and Summary metrics

Sector / Server	Grade	Certificate	Protocol Support	Key Exchange	Cipher Strength
Education / IIM Bangalore	A+	100	100	90	90
Education / SPIT	B	100	70	90	90
Banking / Axis Bank	A+	100	100	90	90
Banking / JSB Pune	B	100	70	90	90
Trading / Upstox	A+	100	100	90	90
Trading / Groww	A+	100	100	90	90

Metric values are read visually from the 0–100 bar charts in the screenshots (about 90 where a bar ends at the 90 mark, 70 for the shortened Protocol Support bar). They are screenshot-derived, not exported test data.

Table 2. Protocol and policy notices

Server	TLS 1.3	PQC key exchange	Long-duration HSTS	Other visible note
IIM Bangalore	Yes	Yes	Yes	—
SPIT	No	No	Yes	Supports TLS 1.0 and 1.1; grade capped to B
Axis Bank	Yes	Yes	Yes	—
JSB Pune	Yes	No	Yes	Supports TLS 1.0 and 1.1, grade capped to B; SNI-only
Upstox	Yes	Yes	Yes	SNI-only
Groww	Yes	Yes	Yes	SNI-only

SECTOR-WISE ANALYSIS

1. Education

IIM Bangalore scored A+ and SPIT scored B. Both show Certificate 100 and about 90 for Key Exchange and Cipher Strength. The gap is in Protocol Support: IIM Bangalore reaches 100, while SPIT drops to about 70 because it still accepts TLS 1.0 and TLS 1.1, which caps its grade at B. SPIT also reports no TLS 1.3 support, whereas IIM Bangalore supports TLS 1.3 and PQC key exchange. Both servers deploy long-duration HSTS. For SPIT, retiring TLS 1.0/1.1 and enabling TLS 1.3 are the clear configuration steps that would remove the cap.

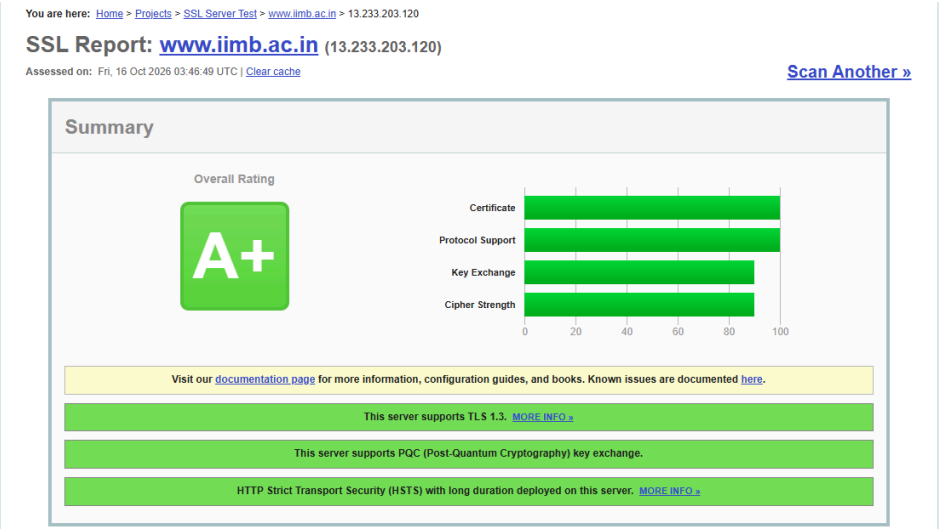


Figure 1. *IIM Bangalore (www.iimb.ac.in) — SSL Labs Summary, overall grade A+.*

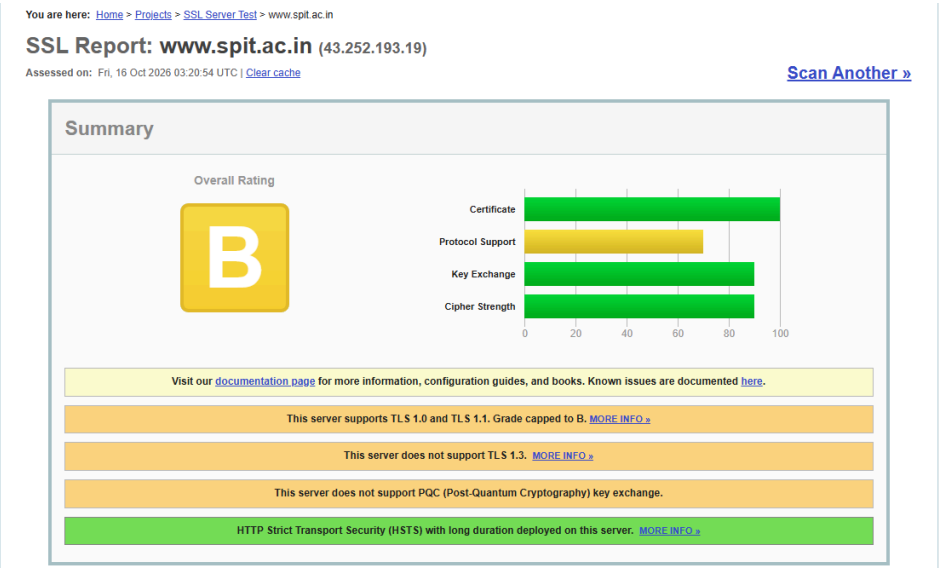


Figure 2. *SPIT Mumbai (www.spit.ac.in) — SSL Labs Summary, overall grade B.*

2. Banking

Axis Bank scored A+ and JSB Pune scored B, the same pattern seen in Education. Both have Certificate 100 and about 90 for Key Exchange and Cipher Strength, both support TLS 1.3 and both deploy long-duration HSTS. Axis reaches Protocol Support 100 and reports PQC key exchange. JSB Pune's Protocol Support is about 70 because TLS 1.0 and 1.1 remain enabled, capping the grade at B, and it reports no PQC support; it is also reachable only by SNI-capable browsers. Notably, JSB Pune does support TLS 1.3, so supporting the newest protocol does not offset keeping the oldest ones enabled.

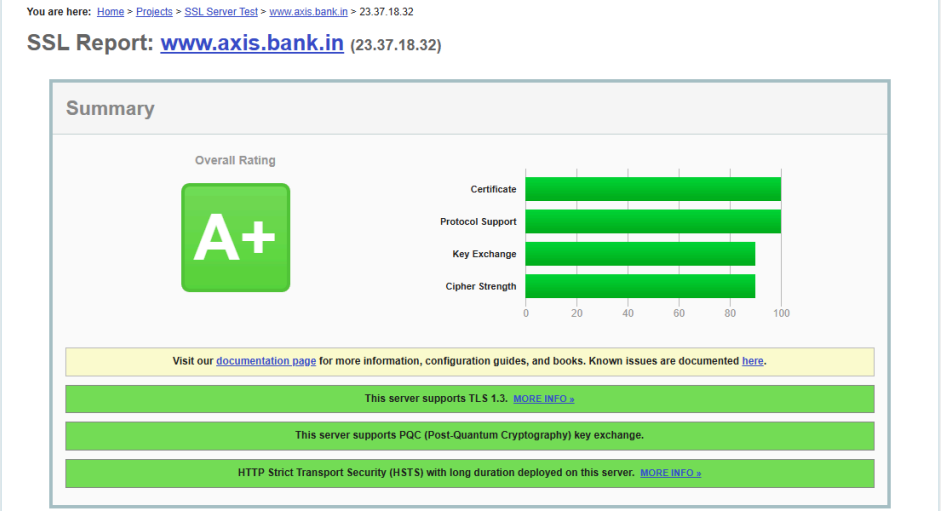


Figure 3. *Axis Bank (www.axis.bank.in) — SSL Labs Summary, overall grade A+.*

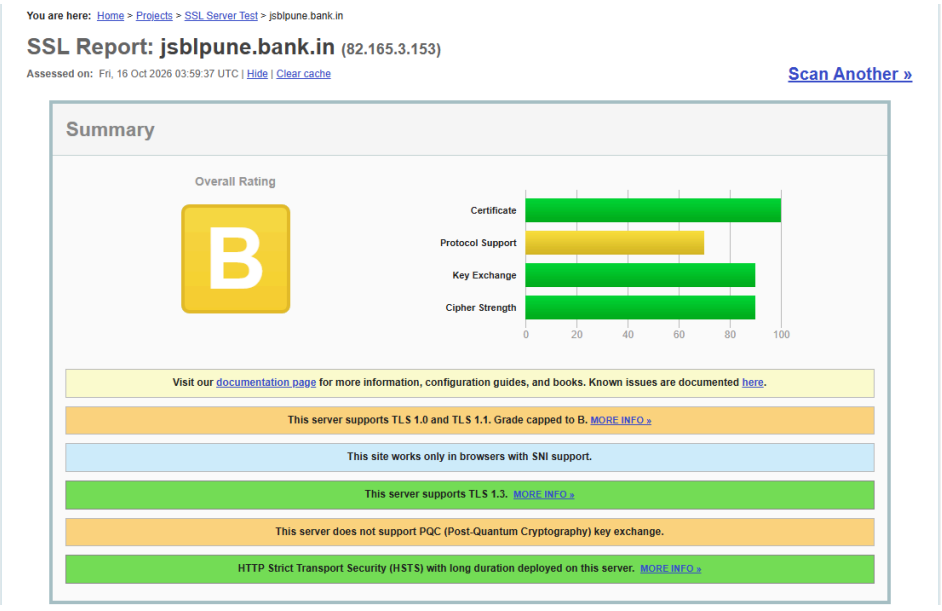


Figure 4. *JSB Pune (jsblpune.bank.in) — SSL Labs Summary, overall grade B.*

3. Trading

Upstox and Groww both scored A+, with identical bars: Certificate 100, Protocol Support 100, Key Exchange about 90 and Cipher Strength about 90. Both support TLS 1.3, PQC key exchange and long-duration HSTS, and both note SNI-only operation. Because the two summaries are effectively identical, the Summary screenshots show no differentiating factor within this sector. Trading is the only sector where both servers earned the top grade.

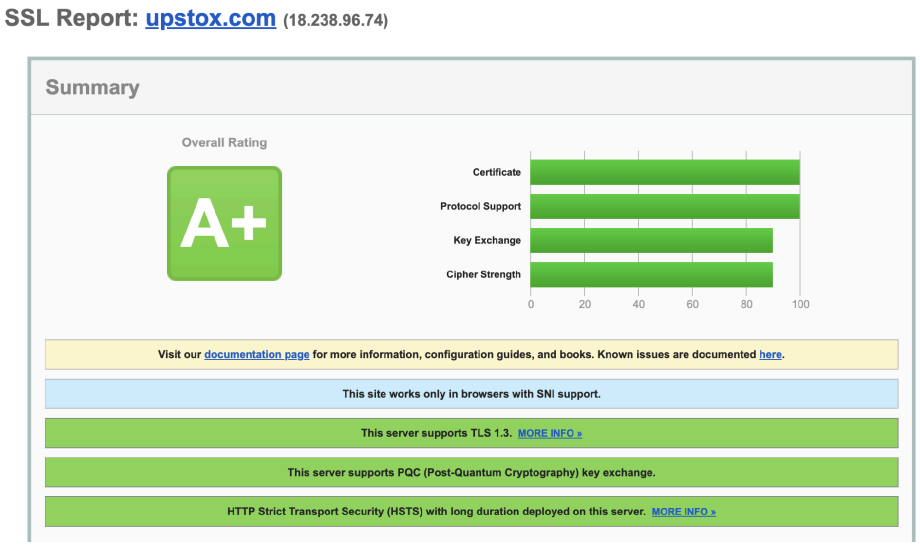


Figure 5. *Upstox (upstox.com) — SSL Labs Summary, overall grade A+.*

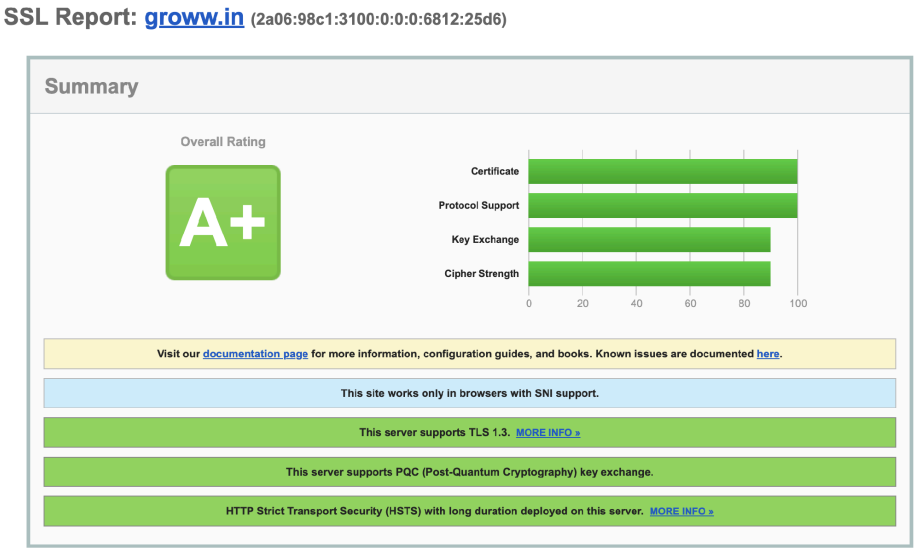


Figure 6. *Groww (groww.in) — SSL Labs Summary, overall grade A+.*

CROSS-SECTOR COMPARISON AND INTERPRETATION

Sector	Best host	Weakest host	Grade spread	Main differentiator
Education	IIM Bangalore	SPIT	A+ / B	TLS 1.0/1.1 enabled; no TLS 1.3, no PQC at SPIT
Banking	Axis Bank	JSB Pune	A+ / B	TLS 1.0/1.1 enabled; no PQC at JSB Pune
Trading	Upstox / Groww	None (tie)	A+ / A+	No difference visible in the Summary

- Four of the six servers earned A+ (IIM Bangalore, Axis Bank, Upstox, Groww) and two earned B (SPIT, JSB Pune).
- Every server scored 100 for Certificate and about 90 for Key Exchange and Cipher Strength, so certificates and cipher strength did not separate the grades.
- Both B grades come from the same cause: TLS 1.0/1.1 is still accepted, which lowers Protocol Support to about 70 and caps the grade.
- Five of six servers support TLS 1.3 (SPIT does not); four support PQC key exchange (IIM Bangalore, Axis, Upstox, Groww); all six deploy long-duration HSTS.

The comparison covers particular hostnames at the time of testing, not a sector-wide audit. Two sites per sector cannot show that one industry is inherently more secure than another.

BROWSER/SERVER HANDSHAKE BEHAVIOUR — INTERPRETATION

The Summary pages show the server capabilities that decide what a browser and server can negotiate, but not a specific browser/OS handshake simulation. A TLS 1.3-capable browser can negotiate TLS 1.3 with IIM Bangalore, Axis Bank, JSB Pune, Upstox and Groww. SPIT does not offer TLS 1.3, so a browser connecting there falls back to TLS 1.2 or, for a very old client, to TLS 1.0/1.1. This fallback is exactly what the B grade penalises. Upstox, Groww and JSB Pune require SNI, so clients that do not send the hostname in the ClientHello cannot complete the handshake. The exact protocol and cipher suite chosen in a given session cannot be determined from the screenshots alone.

Lab 7 item	Evidence available / additional work needed
Summary	Captured for all six hosts; grades and four metric bars documented.
Certificate details	Certificate score shown; issuer, expiry and chain not captured.
Configuration / TLS versions	TLS 1.3 yes/no and TLS 1.0/1.1 status noted; full version list not captured.
Cipher suites / preference	Cipher Strength score shown; individual suites and ordering not captured.
Handshake simulation	Not captured; no negotiated protocol or cipher is claimed.

RESULTS

Six public web servers were compared using Qualys SSL Labs Summary reports. Education: IIM Bangalore A+, SPIT B. Banking: Axis Bank A+, JSB Pune B. Trading: Upstox A+, Groww A+. The two B grades are explained by continued support for TLS 1.0 and TLS 1.1, which lowers Protocol Support to about 70; every other metric was equal across all six servers. TLS 1.3, PQC key exchange and SNI behaviour varied between hosts, while long-duration HSTS was present on all six.

CONCLUSION

Qualys SSL Labs gives a structured way to evaluate public HTTPS configurations. Across the three sectors the grades ranged from B to A+, and the deciding factor was protocol hygiene rather than certificate or cipher strength: servers that retired TLS 1.0/1.1 and enabled TLS 1.3 earned A+, while those still accepting legacy protocols were capped at B. The Trading sector was uniformly strong, whereas Education and Banking each contained one weaker host. The experiment achieved a comparative assessment of server configuration; observing actual negotiated handshakes would additionally require the Handshake Simulation section of the detailed reports.